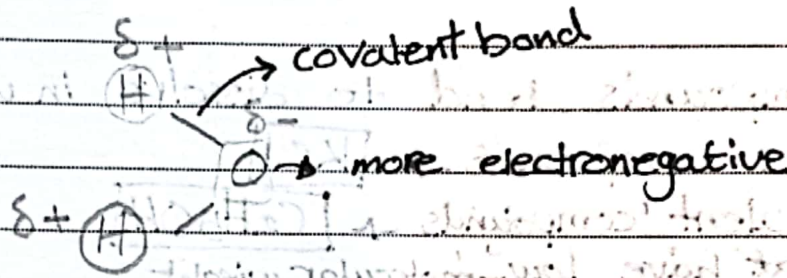


## No. Chapter (1)

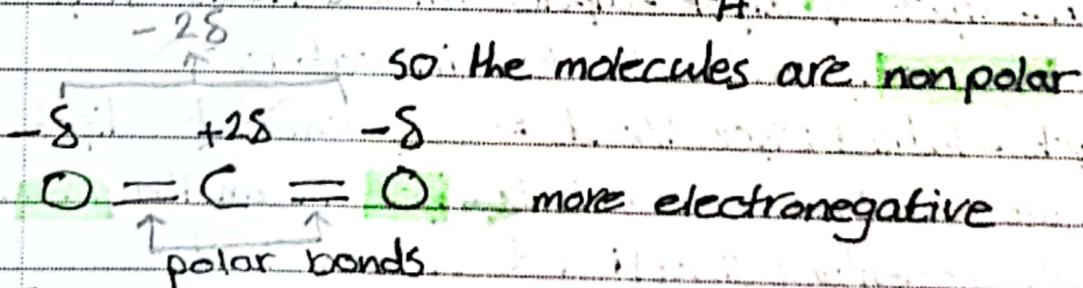
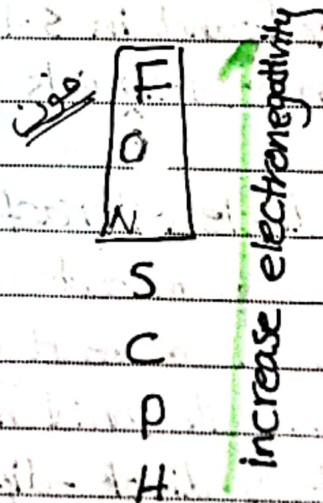
**polar bond:** covalent bond between (2) atoms where the electrons forming the bond are **unequally distributed**, more negative charge found closer to one atom. Due to difference in electronegativity of atoms involved in bond.



**\* Electronegativity:** a measure of the force of an atom's attraction for electrons it shares in a chemical bond with another atom.

**Note 80**

A dipole moment is simply the measure of net polarity in a molecule [bonds with positive & negative ends]



- $[CO_2]$  have polar bonds, but given their geometry are non polar molecules which means that they have a **zero dipole moment**



\* ملاحظة :- إذا بنا ضد إذا المركب عبارة عن dipole moment أو لا يشترك الشحنات الجزيئية الذرات وينجم عن ذلك صفرية وعبارته هو zero dipole moment

⇒ Compounds tend to dissolve in water :-

(1) Ionic compounds →  $KCl$

(2) low-molecular-weight covalent compounds →  $C_2H_5OH$

\* Ion-dipole interaction :-

→ Commonly found in solutions, especially ionic compound dissolved in polar liquids.

\* Dipole-dipole interaction :- when the partial charges formed within one molecule are attracted to an opposite partial charge in a nearby molecule. [the partial positive side of one molecule attracting the partial negative side of another molecule]

• Ion-dipole & dipole-dipole interactions help ionic & polar compounds dissolve in water.

\* Dipole-Induced Dipole Bonds :- permanent dipole in one molecule can induce a transient dipole in another molecule through momentary distortion of electron clouds

\* Dipole - Dipole interactions :-

alcohol  $[ROH]$  & ketone  $[R_2C=O]$

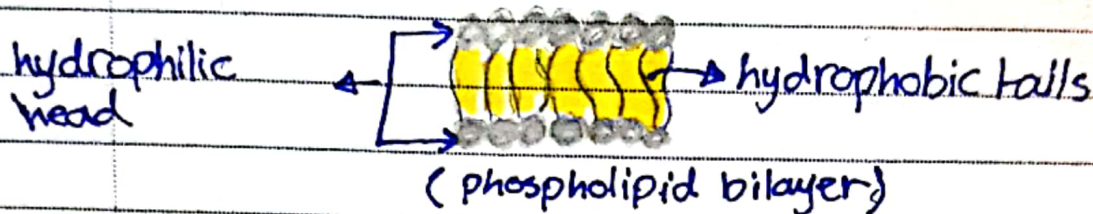
- الجزء الموجب من الكحول (الهيدروجين)  
يلجذب للجزء السالب من الكيتون  
(الأكسجين)

- والأكسجين السالب الي الكحول  
يلجذب للهيدروجين الي الكيتون .



No. Chapter (1)

- **Hydrophilic** → tend to dissolve in water **water-loving**
- **Hydrophobic** → tend to **not** dissolve in water **water-fearing**



- **Amphipathic** → has characteristics of both properties
  - \* molecules that contain one or more hydrophobic & one or more hydrophilic regions → Sodium palmitate
  - ↳ Oils & Fats → Lipids
  - ↳ saturated fatty acid

(tail) ذيل ←  
(head) رأس ←

- Note :**
- \* **saturated fatty acid** → no double bonds
  - \* **unsaturated** " " → there is a double bond

**Van der Waals** Interactions between nonpolar molecules

↳ the weakest intermolecular force

consist of dipole-dipole forces & dispersion forces  
↳ (London)

**Hydrophilic :**

- **Polar covalent compounds**
  - Alcohols
  - Amino acids, phosphate esters
  - Ketones
  - Sugars
  - Ionic compounds

**Hydrophobic :**

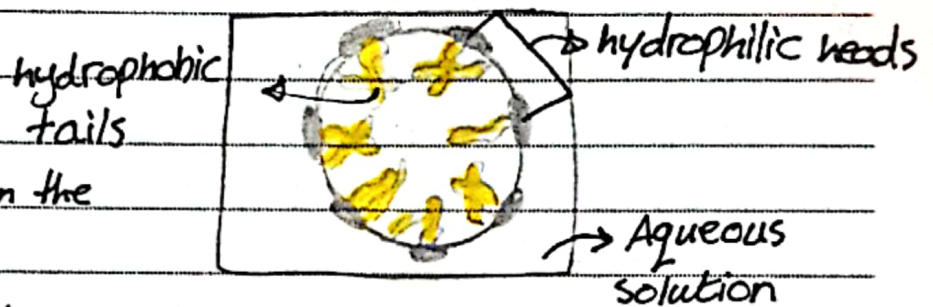
- **Nonpolar covalent compounds**
  - Hydrocarbons [Hexane...]
  - Fatty acids
  - Cholesterol

## ■ Micelle Formation by amphipathic molecules 89

**Micelle:** a spherical arrangement of organic molecules in water solution clustered → composed of phospholipids

→ **Hydrophobic parts:** buried inside the sphere

→ **Hydrophilic parts:** on the surface & in contact with water



\* Formation depends on the attraction between

**temporary induced dipoles**

a weak attraction that results when a polar molecule induces a dipole in an atom or in a nonpolar molecule

**Hydrogen bonds:** non-covalent

— positive end of one dipole is a hydrogen atom bonded to an atom of [High electronegativity O/F/N]

— Negative end of the other dipole is an atom with a lone pair of electrons [O/N]

\* Hydrogen bonds are much weaker than normal covalent bond

\* Each water molecule can be involved in 4 hydrogen bonds: 2 as donor, & 2 as acceptor

Due to the tetrahedral arrangement of the water with bond angle  $104.3^\circ$



## No. Chapter (1)

→ Even though hydrogen bonds are **weaker** than covalent bonds, they have effect on the physical properties of hydrogen bonded compounds

→ The **boiling point** for water is **higher** than  $(NH_3 // CH_4)$  because the attraction between water molecules is high & the molecular weight is higher because of the number & the strength of H-bonds  
→ **Melting point** is the same.

### # The importance of Hydrogen bonding

→ stabilization of 3-D structures of biological molecules such as: DNA // RNA // proteins.

important in protein & nucleic acid structures.

- purines : (2) rings fused together **Guanine // Adenine**

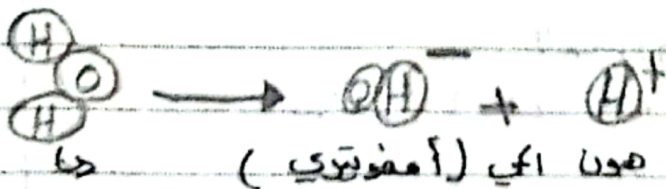
- pyrimidines : simple ring **thymine // cytosine // uracil**

G≡C // A-T

## Acids, Bases & pH

Acid : **Proton donor**

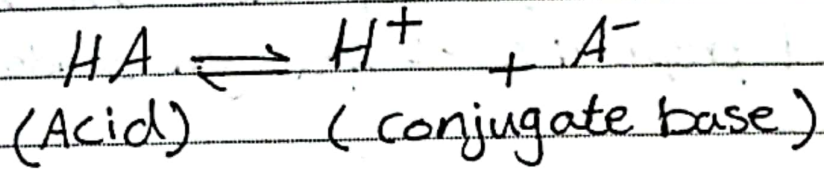
Strong Base : **proton acceptor**



الحماة : عادة تنتج أيون  $(H^+)$  يعني ينتج حمض قوي

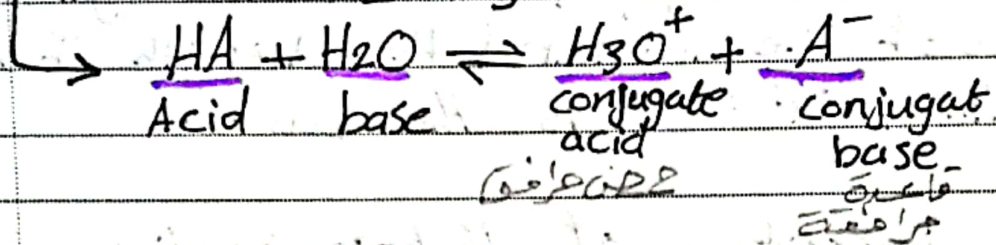
القاعدة : عادة تنتج أيون  $(OH^-)$  يعني يستقبل حمض قوي

## Acid Strength 80



$$K_a = \frac{[H^+][A^-]}{[HA]}$$

الصيغة  
البديلة  
للمفاعل



$K_w$  : ion product constant ثابت حاصل حاصل

$$K_a = \frac{[H^+][OH^-]}{[H_2O]}$$

تركيز الماء ثابت

$$K_w = [H^+] \cdot [OH^-]$$

لحساب  
مركبات ثابت

$$\text{at } 25^\circ \quad [H^+] = [OH^-] = 10^{-7}$$

$$K_w = 10^{-14}$$

بمطابق تركيز أيون  
الهيدروكسجين و أيون الهيدرونيوم

$$pH = -\log_{10} [H^+]$$



No. Chapter (1)

$$\rightarrow K_a = \frac{[H^+].[A^-]}{[HA]} \rightarrow [H^+] \times \frac{[A^-]}{[HA]}$$

$$\log K_a = \log [H^+] + \log \frac{[A^-]}{[HA]}$$

تذكر  
اللوغاريتم في حالة الضرب يتحول إلى جمع

$$-\log [H^+] = -\log [K_a] + \log \frac{[A^-]}{[HA]}$$

$$pH = pK_a + \log \frac{[A^-]}{[HA]}$$

حسب قانون ال (pH)

✱ Henderson-Hasselbalch equation :-

$$pH = pK_a + \log \frac{[\text{conjugate base}]}{[\text{weak acid}]}$$

1] When the concentrations of weak acid & its conjugate base are equal, the pH of the solution equals the pKa of the weak acid.  
في لوغاريتم الواحد = 0

→ when  $pH < pK_a$ , the weak acid predominates

→ when  $pH > pK_a$ , the conjugate base predominates

✱  $pH < pK_a \rightarrow$  protonated form

✱  $pH > pK_a \rightarrow$  deprotonated form



\* **Titration** : an experiment in which measured amounts of acids are added to base

**Equivalence point** :- The point in an acid-base titration at which the amount of acid is neutralized with the amount of base.

- **monoprotic acid** →  $1 H^+ / \text{mole}$  **acetic acid**
- **diprotic acid** →  $2 H^+ / \text{mole}$  **carbonic acid**
- **triprotic acid** →  $3 H^+ / \text{mole}$  **phosphoric acid**

\* **Buffer** : a solution whose pH resists change upon addition of small to moderate amounts of a strong acid or base. // consists of a weak acid & its conjugate base.

⊗ They keep the pH constant by taking up protons which are released during reaction, or by releasing protons when they are consumed by reactions.

← لما نضيف حمضاً عاكساً :-

الـ pH يقل

Buffer : يحاول تقليل من قوة الحمض حتى لا يتأثر الـ (pH)

← لما نضيف قاعدة عاكساً :-

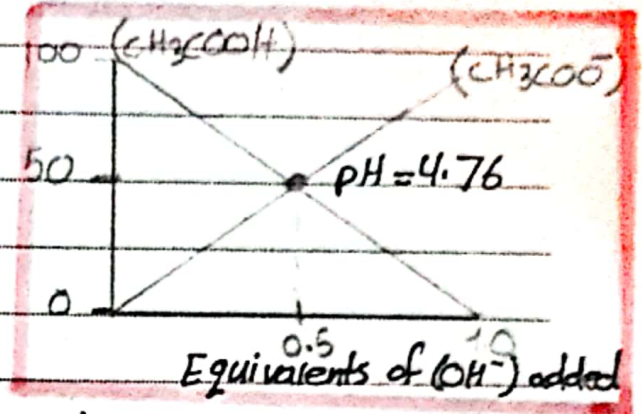
الـ pH يزيد

Buffer : يحاول تقليل من قوة القاعدة حتى لا يتأثر الـ (pH)

## No. Chapter(2)

- Inflection point: the point in the titration at which  $[\text{acid}] = [\text{conjugate base}] \rightarrow \text{pH} = \text{pK}_a$  happens when we add (0.5) equivalence of  $(\text{OH}^-)$

- Starting 100% of acetic at very low pH  $[\text{OH}^- \text{ is } 0]$ , & also here no conjugate base appears, as we start adding the  $(\text{NaOH}) - (\text{OH}^-)$  the acetic acid concentration start to decrease

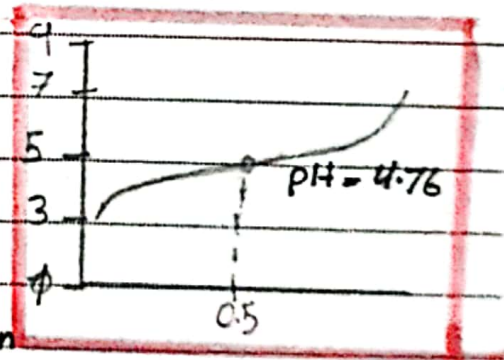


& the acetate [conjugate base] starts appearing

more  $(\text{OH}^-)$  is added  $\rightarrow$  more acetate until we reach point where  $[\text{acetic acid}] = [\text{acetate}]$

more  $(\text{OH}^-)$  is added  $\rightarrow$  more & more acetate until we reach [1.0 equivalence] of  $\text{OH}^-$  at this point the acetic acid is fully deprotonated & its percentage becomes zero while acetate 100%.

nothing new regarding the concentration of acetic acid & acetate, but:



Firstly, there is a rapid increase in (pH) as we add  $(\text{NaOH})$  to the solution

There is a region where (pH) increases slightly as we add more  $(\text{OH}^-)$  (a buffer region)  $\text{pH} = \text{pK}_a \pm 1$

After this region (pH) increase as firstly (rapid increase)



## No. Chapter (2)

\* Buffering capacity is related to the concentrations of the weak acid & its conjugate base

• the greater the concentration of the weak acid & its conjugate base, the greater the buffer capacity.

→  $\text{H}_2\text{PO}_4^- / \text{HPO}_4^{2-}$  is the principal buffer in cells

→  $\text{pK}_a$  of  $\text{H}_2\text{CO}_3$  is (6.37), The pH of blood (7.4)

→  $\text{H}_2\text{CO}_3 / \text{HCO}_3^-$  is an important [but not the only] buffer in blood.

\* Hyper ventilation:- can result in increased blood (pH)

\* Hypo ventilation:- can result in decreased blood (pH)

⇒ Selecting a Buffer:-

• Suitable  $\text{pK}_a$

• no interference with the reaction or detection of the assay

• suitable ionic strength

• suitable solubility

• its non-biological nature

• Zwitterions: compounds that have both a positive & negative charge.